MILITARY SPECIFICATION

CARTRIDGE, 9MM, BALL, REFERENCE

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Cartridge, 9MM, Ball, Reference to be used for calibration purposes in 9mm cartridge acceptance testing equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extend specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105- Sampling Procedures and Table for Inspections by Attributes

MIL-STD-636- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-644- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168- Lot Numbering of Ammunition
DOD-STD-1468- Small Caliber Ammunition Test Procedures, 9MM Cartridges

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT & ENGINEERING CENTER (ARDEC)

12556029	-	Cartridge, 9MM, Ball, Reference
7643674	-	Classification of Cartridge Case Defects for Small Arms Ammunition
9396206	-	Packing and Marking for Box, Wirebound, for Cartridges, Small Caliber
LI12556029-		Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting "officer".)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1. Parts. The parts shall comply with all requirements specified on Dwg. 12556029, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Bullet extraction. The average force necessary to extract the bullet from the cartridge case shall be not less than 80 lbs, nor shall any individual extraction force be less than 60 lbs.

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3.4 Residual stress. The cartridge case shall not split when subjected to one percent mercurous nitrate solution for 15 minutes.

3.5 Velocity. The average velocity of the cartridges when conditioned at 21° ± 2° Centigrade (C) shall be 375 meters per second (m/sec) plus or minus 15 m/sec at a point 16 meters from the muzzle. The standard deviation of the velocities shall not exceed 5 m/sec.

3.6 Chamber pressure. The average peak chamber pressure of the cartridge at the case mouth position when conditioned at 21°C ± 2°C shall not exceed 215 Megapascals (MPa) and no individual peak pressure shall exceed 250 MPa. The standard deviation of the sample cartridges shall not exceed 10 MPa.

3.7 Function and casualty. The cartridge shall function without casualty.

3.8 Waterproof. When placed under water and subjected to a one half atmosphere negative pressure for 30 seconds, the cartridge shall not release more than one bubble of air.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.9.2 Foreign matter. The cartridge shall be free of corrosion, stains, discoloration, dirt, oil, and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements. The provisions of MIL-A-48078 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or

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supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality Conformance Inspection (See 4.3)

4.3 Quality conformance inspection.

4.3.1 Lot formation. In accordance with MIL-A-48078.

4.3.1.1 Lot submission. The product shall be submitted in accordance with MIL-STD-105. The entire lot of reference cartridges shall be assembled and charged on a single loading machine (see 6.6).

4.3.1.2 Component parts. Unless otherwise specified, component lots shall be homogeneous and of a size convenient to the contractor and inspected, tested and accepted by the contractor. The cartridge lot shall contain:

a. Cartridge cases from one lot interfix number and one manufacturer.

b. Primers from one lot and one manufacturer.

c. Bullets from one lot interfix number and one manufacturer.

d. Propellant from one lot and one manufacturer.

4.3.1.3 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, as well as a lot number in accordance with MIL-STD-1168.

4.3.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Specified AQL's for major and minor defects shall be applied on a class basis.

b. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot

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shall then be rescreened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition, the portion of the lot remaining to be packed shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70621(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Cartridge, 9MM, Ball Reference	1 OF 3	12556029
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1.	Split case in K, L, or M location (See Dwg. 7643674)		100%	3.2	Visual
2.	Perforated Case		100%	3.2	Visual
3.	No primer		100%	3.2	Visual
4.	Cocked primer		100%	3.2	Visual
5.	Inverted primer		100%	3.2	Visual
MAJOR:			0.25%		
101.	Cartridge, corroded or stained, if etched			3.2	Visual
102.	Case, round head			3.2	Visual
103.	Split case in I or J location (See Dwg. 7643674)			3.2	Visual
104.	Case, beveled underside of head			3.2	Visual
105.	No radius on head (rim)			3.2	Visual
106.	Split bullet jacket			3.2	Visual
107.	Loose bullet			3.2	Visual
108.	Loose primer			3.2	Visual
109.	Total length of cartridge, incorrect			3.2	Gage
110.	Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile & alignment gage)			3.2	Gage

NOTES: Refer to MIL-STD-636 (caliber .45 section) for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70621(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 12556029
4.3.2.1	Cartridge, 9MM, Reference				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111.	Diameter of extractor groove, incorrect			3.2	Gage
112.	Diameter of head, incorrect			3.2	Gage
113.	Thickness of head, incorrect			3.2	Gage
114.	Depth of primer, incorrect			3.2	Gage
115.	Case length, incorrect			3.2	Gage
MINOR:			1.5%		
201.	Cartridge, discolored, dirty, oily smeared			3.2	Visual
202.	Case dent			3.2	Visual
203.	Case draw scratch			3.2	Visual
204.	Case scratch			3.2	Visual
205.	Case scaly metal			3.2	Visual
206.	Case fold			3.2	Visual
207.	Case wrinkle			3.2	Visual
208.	Case buckle			3.2	Vusual
209.	Case bulge			3.2	Visual
210.	Illegible or missing head stamp			3.2	Visual
211.	Defective head			3.2	Visual
212.	Defective mouth			3.2	Visual
213.	Bullet dent			3.2	Visual
214.	Bullet scratch			3.2	Visual
215.	Scaly metal (bullet)			3.2	Visual
216.	Upset (crooked) point (bullet)			3.2	Visual

NOTES: Refer to MIL-STD-636 (caliber .45 section) for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70621(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 12556029
4.3.2.1	Cartridge, 9MM, Reference				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
217.	Flat point (bullet)			3.2	Visual
218.	Nicked or dented primer			3.2	Visual
219.	No waterproofing material (primer pocket joint)			3.2	Visual
220.	Workmanship			3.9	Visual

NOTES: Refer to MIL-STD-636 (caliber .45 section) for visual defects. In the event of conflict between paragraphs of this specification and MIL-STD-636 as to defect classification, the classification specified herein shall apply.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Tests. The tests listed in Table I shall be conducted in accordance with the methods and procedures specified in 4.5.

4.4.3.1 Test samples. The quantities for the various tests shall be as specified in Table I. The test samples shall be selected from lots that have met the visual and dimensional inspection requirement.

TABLE I. Tests.

TEST	NUMBER OF CARTRIDGE
Bullet extraction 1/	20
Residual stress 1/	50
Velocity 2/, 3/	300
Chamber pressure 2/, 3/	(300)
Function & Casualty 4/	(300)
Waterproof 1/	20

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot, subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

2/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot.

3/ Velocity and chamber pressure tests shall be conducted concurrently.

4/ Function and casualty testing shall be performed concurrently with velocity and pressure testing. The lot shall be rejected when firing defects exceed the acceptance numbers in Table II.

4.4.4 Packaging, packing, and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than high pressure test shall be classed as a major defect. All non-conforming cartridges shall be rejected.

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4.4.5 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Paragraph Reference/Inspection method column in the tables starting with Paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5.1 Index of inspection equipment list. The examination tests shall be made using equipment listed on LI12556029.

TABLE II. Firing defects.

DEFECTS	ACCEPTANCE NUMBERS
(a) Misfire	1
(b) Bullet remaining in bore	0
(c) Perforation in firing pin indent in primer cup	1
(d) Escape of gas around primer cap	
(i) more than 50% of periphery	3
(ii) less than 50% of periphery	6
(e) Blown primer (1)	0
(f) Loose primer which falls out of pocket	0
(g) Loose primer which does not fall out of pocket	2
(h) Case split (longitudinal)	
(i) Mouth (defined as the length of bullet intrusion)	3
(ii) Body	1
(iii) Through the head	0
(j) Case rupture (circumferential)	
(i) Partial rupture	1
(ii) Complete rupture	0

(1) Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted.

4.5 Methods of inspection.

4.5.1 Bullet extraction. The test shall be conducted in accordance with DOD-STD-1468.

4.5.2 Residual stress (mercurous nitrate). The test shall be conducted in accordance with DOD-STD-1468.

4.5.3 Velocity. The test shall be conducted in accordance with DOD-STD-1468.

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4.5.4 Chamber pressure. The test shall be conducted in accordance with DOD-STD-1468.

4.5.5 Waterproof. The test shall be conducted in accordance with DOD-STD-1468, Type II Test.

4.5.6 Defect penalty. In any ballistic test in which the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table II.

4.5.7 Test validity. If for any reason the testing facility considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

5. PACKAGING

5.1 Packing. Level A (World Wide Shipment). The cartridges shall be packed in accordance with Drawing 9396206.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use as reference cartridges for calibration purposes in 9mm cartridge acceptance testing.

6.2. Ordering data. Ordering data should include:

a. Title, number and date of specification.

b. Type and level of packing for the cartridges

c. Provisions for the submission of Inspection Equipment Designs (see 6.3).

d. Provisions for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I(D) Dover, NJ 07801-5001.

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6.4 Hazard notice. The cartridges described herein and some of its components are flammable and explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, and electrically inducted incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development & Engineering Center (ARDEC) may also include drawing prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 In-process controls.

6.6.1 Uniformity of charge. A machine shall be chosen which is known to drop charges with better than average uniformity and which gives constant results in the daily production tests. If relative machine performance is unknown, special tests should be fired to assure selection of a good machine.

6.6.2 Loading control. A quantity of 20 rounds shall be fired for pressure and velocity for each four hours of an eight hour shift to establish velocity and pressure values, and check charge weight and uniformity of charge weight every hour. Once established, the charge weight shall remain constant throughout loading.

6.7 Subject term (key word) listing.

Cartridge, 9mm
Military Specification
Reference Cartridge
Small Caliber Ammunition

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AC05)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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